Multi-modal Sentiment Analysis for Yelp Reviews: A Conflict-Aware Gating Mechanism

Group 22
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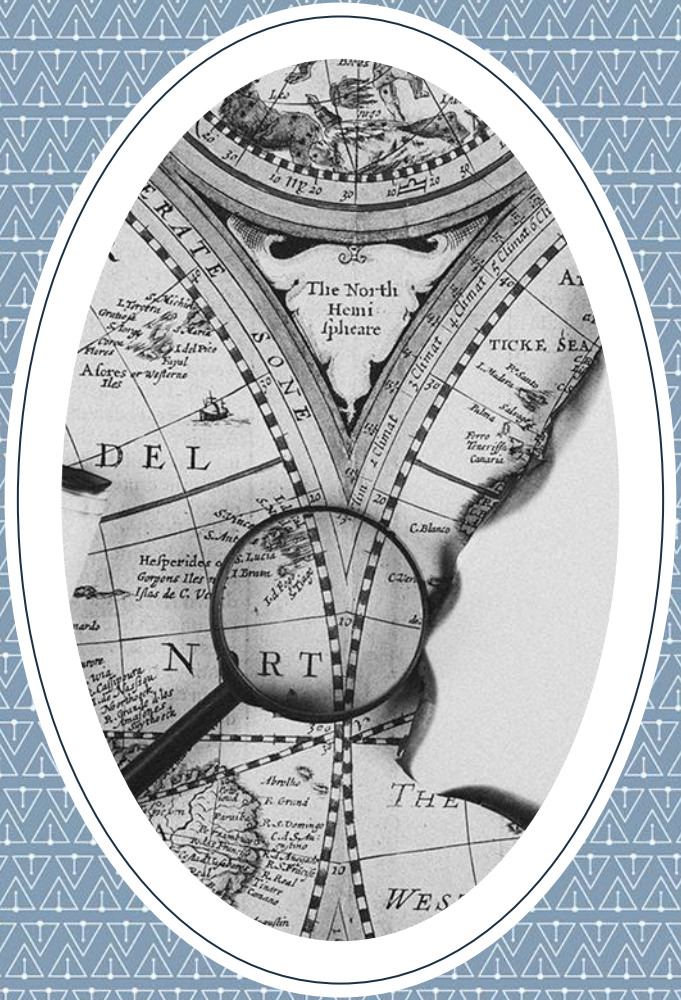
01

Introduction

Overview of the problem domain, the motivation behind the research



Background

Features of online review platforms (like Yelp):

Multi-Modality: Users often provide multiple types of information

- Textual reviews
- Visual images
- Numerical ratings

etc.

Cross-modal sentiment conflicts: It is common that different forms of expression convey contradictory sentiments

Background

Current Models

- Uniform fusion
- No explicit conflict modeling

Results in:

- Biased predictions
- Limited interpretability

Our Goal

- Addressing modality conflicts explicitly

Aims to:

- Build more robust models
- Maintain performance even facing with inconsistency

Current Research Gap

Single-modality models: Lack cross-modal validation

Traditional fusion: Suffers from modality imbalance

Existing gated models:

- Rely on:
  - Static confidence
  - Independent reliability scores
- But Missing:
  - Dynamic conflict quantification
  - Joint modeling of similarity and polarity divergence

02

Proposed Approach

Presentation of our novel solution, the technical design of the model, as well as the dataset used

Core Contribution

We introduce CMAG (Conflict-Aware Gating Mechanism), which:

- Measures cross-modal feature similarity
- Computes sentiment polarity divergence
- Quantifies conflict intensity
- Dynamically generates fusion weights

Dataset

Name: Yelp Dataset

Introduction: A subset of Yelp's businesses, reviews, and user data

Quantity: ~10,000 multimodal reviews

Data processing:

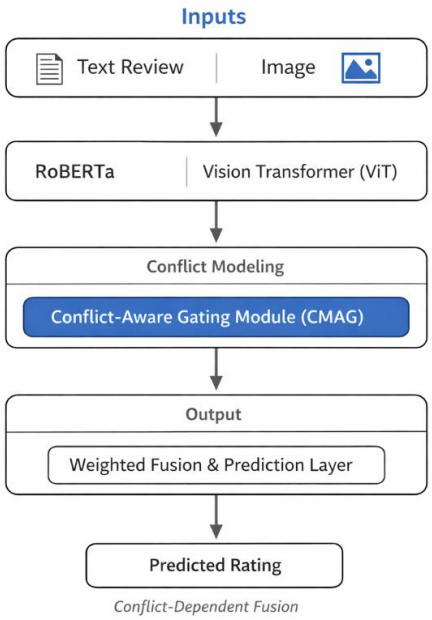
- Extract polarity and confidence scores
- Identify conflicting samples
- Conduct exploratory data analysis

Key Statistics:

- Average values
- Value ranges
- Distributions

Project Design

- Input: Multi-Modal Data
- Text encoder: RoBERTa
- Image encoder: ViT
- Conflict modeling: CMAG
- Output: Weighted fusion + prediction layer

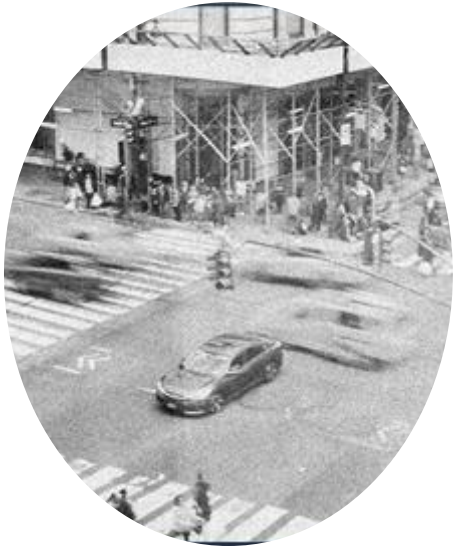


03

Implementation & Evaluation

Details of the model implementation, the evaluation strategy, and the expected outcomes

Implementation



Language: Python 3

Core Framework: PyTorch

Key components:

- RoBERTa and ViT encoders
- CMAG module
- Training pipeline

Training Procedure:

- Baseline comparisons
- Hyperparameter tuning
- Evaluation on validation sets

Evaluation Strategy

A two-fold evaluation

Predictive performance:

- Accuracy / MAE / F1
- Baseline comparison
- Conflict-heavy subset evaluation

Interpretability:

- Modality weight visualization
- Trust map analysis

Expected Contributions

- Novel conflict-aware multimodal framework
- Improved performance on conflicting samples
- Enhanced interpretability
- New insights for online review sentiment modeling

04

Plans & Team

Timeline and team organization

Timeline

Week 6

- Proposal preparation
- Initiation of data preprocessing

Week 7

- Model design based on the pre-processed data

Week 8

- Model training
- Preliminary performance evaluation

Week 9

- Continued model training
- In-depth performance assessment

Timeline

Week 10

- Conflict Analysis and interpretation of modality interactions

Week 11

- Report writing based on experimental results and findings

Week 12

- Model refinement
- Report revision
- Supplementary experiments as necessary

Week 13

- Final submission of project deliverables

Team Orginazation

Data Analysiss

Deep Learning Engineering

Visualization

Evaluation

Thank you!